

Research Article

# Wormhole Detection Scheme With Adaptive Deep Neural Network And Hybrid Multi-Objective Mitigation For Internet Of Things

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## Abstract

The Internet of Things (IoT) is considered a significant innovation by researchers and industry leaders, offering significant advancements in daily life through the use of smart objects that communicate with one another over the internet. Moreover, because IoT devices are resource-constrained and low-cost, implementing an effective security model has emerged as a major technical challenge in recent years. Among the security threats to wireless networks, the wormhole attack is a particularly important and vulnerable issue, as it can disrupt various routing protocols through its activation in various modes. In the transmission of data from a source node to a destination node in network, compromised nodes, also known as wormhole nodes or attackers, may leak or alter the routing information of a specific path. These paths, which contain malicious nodes, are referred to as wormhole paths. To enhance security in IoT, it is crucial to identify and secure such paths. This research presents a novel model aimed at detecting and preventing wormhole attacks to ensure the secure transmission of data in IoT networks. The proposed model comprises two main phases: (a) the detection phase which utilizes an adaptive Deep Neural Network (DNN) based on the Crow Search with Bird Swarm Optimization (CS-BSO) algorithm to detect wormhole attacks across k-different paths, and (b) the prevention phase which utilizes the CS-BSO algorithm in combination with the Bird Swarm Algorithm (BSA) and Crow Search Algorithm (CSA) to determine the optimal path for data transmission. This optimal path selection is based on a multi-objective function that considers network parameters such as delay or latency, energy, packet loss ratio, distance, and throughput, effectively protecting the network from wormhole attacks.

**Keywords-** Internet of Things; Wormhole Attack Detection and Mitigation; Adaptive Deep Neural Network; Crow Search with Bird Swarm Optimization; Multi-Objective Function

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## I. INTRODUCTION

According to recent projections, the expansion of the IoT represents a new aspect of modernisation. The number of varied IoT devices that are planned to be deployed is estimated to reach 24 billion by the year 2020. Broadly defined, an IoT infrastructure is characterized by the deployment of heterogeneous systems, methodologies, and TCP/IP protocols, all of which are designed to facilitate device-to-device communication and to enable the contextualization of environmental data [9]. "Low-power and Lossy Network (LLN)" is considered as a major example for IoT networks, which includes heterogeneous and tiny devices along with limited processing resources, memory, and power. These networks have been employed in a wider range of applications like water management, urban sensor networks, industrial smart grid systems, health care, smart cities, smart home, and industrial smart grid systems [9] [10]. Though, the utilization of IoT devices strictly focuses on resource constraints in terms of memory, high communication loss rates of the communication links, and energy [11]. In diverse applications, the relationship of network elements is facilitated by routing strategy of IoT network. The standard protocol in IoT is RPL [12] protocol that offers less security against diverse routing attacks [13]. Specifically, the flaws in the design of the configuration operations of the RPL make the network vulnerable to a variety of assaults, including rank, black hole, Sybil, and wormhole attacks. These attacks influence the basic security services like authenticity, access control, data integrity, and confidentiality [14], and would be utilized based on an adversary for running more powerful attacks. The implementation of the data routing in these networks is done to adapt real time applications [15].

One of the primary risks associated with IoT pertains to the wormhole attack. This refers to a scenario in which a rogue node intercepts packets originating from a particular location within the network and subsequently employs a tunnel to establish contact with another malevolent node located elsewhere in the network [16]. Henceforth, the malevolent nodes establish dynamic interconnections, thus facilitating accelerated communication as compared to the other nodes within the network. It is imperative to note, however, that the aforementioned communication requires specialized hardware for successful implementation. The attackers employ wormhole tunnels, which utilize high-quality or wired-link wireless out-of-band links, to carry out their activities [17]. The created tunnel is employed by malicious nodes to make DoS attack. Cryptographic approaches are employed to defend the network transmissions; however, they fail to detect the wormhole attacks owing to independent nature of wormhole attack in the cryptographic method [18] [19]. Moreover, a wormhole attack cannot be detected easily because of the usage of the limited number of resources. The wormhole attack can be established by attackers without revealing their identities [20]. The routing protocols, namely DSR and AODV, are vulnerable to this particular attack.

IoT networks are exposed to diverse attacks like wormhole attack, which is a more powerful attack and prevention of this attack has established as a complex task [21]. A vital assignment of the wormhole attack can indicate a specific breakdown in the transmission among wireless network. In this scenario, a higher-level virtual tunnel is established by two or more malicious network nodes that are conspiring together. This tunnel is then exploited for the transit of packets between the ends of the tunnel [22]. It emulates the shorter links in the network and hence acts as a benefit to unsuspecting network nodes through default search for shorter routes. Diverse prevention approaches like DelPHI is used to prevent the use of AODV routing protocol, innovations diffusion, accept instances of RREQ (Route Request) and recording transfer, etc. Due to the combination of ad hoc networks with pervasive computing environment, the security is the major need. The security is enhanced by applying distributed node behavior monitoring [23]. In [24], security enhanced AODV (AODV-S) is recommended for the purpose of allowing a token to be used by the node prior to the node entering the activities of the network. In existing approaches, diverse protocols use digital signatures or hash chains for protecting the authenticity and integrity of routing information. Further, the SAODV protocol (Secure AODV) is introduced by adopting AODV-S and DelPHI techniques. Similarly, SEAD (Secure Efficient AODV) and Ariadne utilize a variation of the Timed Efficient Stream Loss-tolerant Authentication (TESLA) for achieving the authentication [24]. IDS (Intrusion Detection Systems) have been employed as the subsequent line of security for protecting ad hoc networks [25]. However, the challenges in these approaches should be overcome for preventing the wormhole attacks to enhance the security.

The significant works of the developed wormhole attack mitigation are listed here.

- The aim is to design a secure Internet of Things (IoT) network utilizing an adaptive Deep Neural Network (DNN) that employs the proposed CS-BSO algorithm to detect and mitigate wormhole attacks. The approach involves selecting the optimal path for data transmission to ensure the network's safety and integrity.
- To perform the detection process of wormhole attack by proposed ADNN by considering the input as "source ID, destination ID, RTT, and hop-count" of each node. Here, the proposed CS-BSO algorithm enhances the training of DNN, whose weight is optimized for reducing the error difference among the actual and measured outcomes.
- In order to address the potential threat of wormhole attacks in the context of IoT, the proposed CS-BSO technique employs optimal path selection while taking into account a multi-objective function that includes factors such as distance, energy consumption, transmission delay, packet loss rate, and throughput. This approach aims to effectively mitigate the risk of wormhole attacks in IoT systems.
- To propose a new algorithm called CS-BSO for the enhancing the performance of DNN and for getting optimal

paths from the generated k-paths to detect and mitigate the wormhole attacks.

- To evaluate the efficacy of the constructed model for detecting and mitigating wormhole attacks, several performance metrics will be assessed, including normalized energy consumption, convergence analysis, number of active nodes, and valuable convergence analysis.

The following sections of the proposed model for detecting and mitigating wormhole attacks are presented herein. In Section II, a literature survey is provided. Section IV details the detection of wormhole attacks in IoT using the ADNN model. Section V outlines the mitigation of wormhole attacks in IoT through the use of the proposed CS-BSO model. Section VI delves into the discussion of results. Finally, Section VII concludes this paper.

## II. LITERATURE SURVEY

### A. Related Works

In 2016, Singh *et al.* [1] have developed a Wormhole Resistant Hybrid Technique (WRHT) for detecting the existence of wormhole attack in optimal way than other traditional approaches. It was implemented by using the concept of Delphi and watchdog strategies to ensure the remaining wormhole attacks in the network. It has employed a dual wormhole detection scheme to calculate the probability factors like packet loss probability and time delay probability of the recognized path for finding the probability of wormhole presence value. The nodes along the pathway were assessed according to their behavior, with a focus on the primary feature of Wireless Routing based on Hash Tree (WRHT) - namely, its capacity to protect against all forms of wormhole attacks without necessitating additional hardware such as synchronized clocks, timing information, cryptographic methodologies that demand significant computational resources, or global positioning systems. Empirical findings demonstrate the enhanced efficacy of the proposed model compared to various other approaches to detecting wormhole attacks.

In 2019, Kumar and Sinha [2] have developed approaches like "AODV, FA-AODV, BH-AODV and WH-AODV routing protocol" for simulating the attack in diverse cases. Analyzed through various routing protocols, the proposed model was evaluated in terms of key performance parameters including throughput, End-to-End delay, and Packet Delivery Ratio (PDR). The results of the experiments revealed a decrease in PDR for the FA-AODV protocol, as compared to the WH-AODV and BH-AODV protocols.

In 2020, Kaliyar *et al.* [3] have offered a brief depiction of two main threats for RPL named wormhole and sybil attacks in RPL-based IoT networks. Further, two solutions were developed for detecting these attacks in networks. In particular, this developed approach has employed the theory of Highest Rank Common Ancestor (HRCA) for finding a general ancestor through the highest rank. These approaches have detected ongoing attack and also localized the location of the adversary in the network, which was implemented on network emulators like Contiki and Cooja. It has established the mitigation procedure as fast and lightweight processes. The quantitative results have demonstrated the viability of the

proposed model with regard to network overhead, memory usage, detection time, PDR, and TPR (True Positive Rate).

In 2017, Padmanabhan and Manickavasagam [4] have implemented a distributed and scalable model by using "sequential probability ratio test" for handling high mobility and for avoiding the single point failures without additional resource constraints. It has noted that wormholes have been detected by less number of packets with faster mobility. This model was extremely customizable by selecting the parameters for balancing the accuracy and feed of detection. The proposed model was also analyzed based on constraints like storage, computation, and communication aspects.

In 2020, Rufai *et al.* [5] have offered a "lightweight multi-hop routing protocol in 802.15.4 WSN", which was intended for minimizing the consumption of energy and also for detecting the wormhole attacks. The experimentation outcomes have proven that this implemented MCRP (MAC Centralized Routing Protocol) has established the better performance than other protocols. When the performance of WSN was increased, the node count was increased. Thus, the energy consumption was also reduced.

In 2020, Ali *et al.* [6] have developed a lightweight approach in Mobile ad hoc network (MANET), which was aimed at detecting the wormhole attacks. The present approach entails the calculation of the mean sequence number of reply (RREP) packets transmitted from the source node. The identification of malicious nodes is based on the mean sequence number, which is computed by excluding all traffic. The model exhibits characteristics such as enhanced power efficiency and prolonged network lifespan due to the increased number of data packets transmitted to the destination node. The proposed model has undergone an analysis and comparison against conventional approaches to ascertain its performance.

In 2020, Tamilarasi and Santhi [7] have offered a wormhole attack detection model with secure or optimal path selection. Firstly, multiple or 'K' paths were produced among source and destination nodes using a new protocol called "AOMDV (Ad-hoc on demand Multipath Distance Vector)". Further, the attacked path through wormhole attack was recognized based on the verification of feedback packet and detection packet from the destination. After detecting wormhole-attacked paths by the source node using the "Particle Swarm Optimization (PSO)" algorithm, the optimal path was chosen. The simulation results indicate that the proposed model outperformed traditional methods in terms of improving network lifetime and energy efficiency.

In 2018, Vo *et al.* [8] have implemented a new protocol called MLAMAN (multi-level authentication model) to detect and prevent the wormhole attacks at reliable manner. It was also facilitated the entire intermediate nodes for authenticating the control packets with three levels like integrity verification of packet level, certified the node membership level, and determination of neighborhood correlation among nodes, which was based on a hop-by-hop basis. It was detected that the wormhole nodes was efficiently performed in numerous cases like speeds of moving nodes and variable tunnel lengths. The experimental outcomes have proved the efficiency with different cases.

## B. Problem Statement

During a wormhole attack, an adversary intercepts packets at one point in the network and tunnels them to another location to retransmit them into the network. However, to mitigate the potential damage caused by malicious nodes, it is crucial to monitor the delay in multiple paths to the receiver to ensure the proper functioning of the network. These types of issues must be resolved. Table 1 displays a number of features along with their associated challenges pertaining to the detection and prevention models of wormhole attacks. WRHT [1] can be used in the sensor networks in a direct manner and impractical sensor network assumptions or extra hardware is not needed. But, the wormhole tunnel count is not improved for distinct performance parameters. Novel framework [2] simulates the attacks in various scenarios and several routing protocols like WH-AODV, BH-AODV, FA-AODV, and AODV are implemented. Still, the delays of several paths are not considered by the receiver. LiDL [3] localizes the position of the adversary and the detection approaches are efficient, scalable, and lightweight. Yet, multiple attacks are not prevented and recognized and the solutions for various networking attacks are not examined in the RPL-oriented IoT networks. Scalable and distributed scheme [4] handles the

accuracy as well as speed of the detection and the error rates of the system are also examined. But, the attack is not detected by means of specific validation and application defences. MCRP [5] reduces the energy consumption and the WSN performance is enhanced even when the node count increases. But, confidentiality and authenticity is lacked. Lightweight technique [6] improves network longevity and conservation of node battery power are achieved through the utilization of this approach. However, during the transmission phase, energy consumption is not taken into account and detection of malicious nodes is not accomplished through integration with machine learning methods.. PSO [7] improves the network security and better network lifetime and energy consumption is returned. But, it does not function outside the out-of-band channel. MLAMAN [8] The network employs a management protocol to certify node membership, thereby preventing the infiltration of malicious nodes that attempt to gain access through deceptive means and false information. Still, the practical scenarios that involve the speed of the node and size of the network are not considered and the TLS library does not improve the security performance. Therefore, these challenges serve as a fundamental basis for the development of innovative prevention techniques aimed at bolstering network security.

**TABLE I. REVIEW OF STATE-OF-THE-ART WORMHOLE ATTACK DETECTION AND PREVENTION METHODS**

| Author [citation]                  | Methodology                     | Features                                                                                                                                                                                                                           | Challenges                                                                                                                                                                                                                        |
|------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Singh <i>et al.</i> [1]            | WRHT                            | <ul style="list-style-type: none"> <li>It can be employed in the sensor networks in a direct manner.</li> <li>It does not need any unrealistic sensor network assumptions or extra hardware.</li> </ul>                            | <ul style="list-style-type: none"> <li>It does not enhance the wormhole tunnel count for distinct performance parameters.</li> </ul>                                                                                              |
| Kumar and Sinha [2]                | Novel framework                 | <ul style="list-style-type: none"> <li>Various routing protocols such as WH-AODV, BH-AODV, FA-AODV, and AODV are implemented.</li> <li>The attacks are simulated in various scenarios.</li> </ul>                                  | <ul style="list-style-type: none"> <li>It does not consider the delays of several paths to the receiver.</li> </ul>                                                                                                               |
| Kaliyar <i>et al.</i> [3]          | LiDL                            | <ul style="list-style-type: none"> <li>The detection approaches are efficient, scalable, and lightweight.</li> <li>It facilitates the identification of the adversary's location within the network.</li> </ul>                    | <ul style="list-style-type: none"> <li>It does not recognize and prevent multiple attacks.</li> </ul>                                                                                                                             |
| Padmanabhan and Manickavasagam [4] | Scalable and distributed scheme | <ul style="list-style-type: none"> <li>It investigates the error rates of the system.</li> <li>It balances the accuracy and speed of the detection.</li> </ul>                                                                     | <ul style="list-style-type: none"> <li>It does not detect the attack using specific validation and application defences.</li> </ul>                                                                                               |
| Rufai <i>et al.</i> [5]            | MCRP                            | <ul style="list-style-type: none"> <li>It enhances the WSN performance even when the node count increases.</li> <li>The energy consumption is reduced.</li> </ul>                                                                  | <ul style="list-style-type: none"> <li>It lacks confidentiality and authenticity.</li> </ul>                                                                                                                                      |
| Ali <i>et al.</i> [6]              | Lightweight technique           | <ul style="list-style-type: none"> <li>The battery of the nodes is saved.</li> <li>It enhances the network lifetime.</li> </ul>                                                                                                    | <ul style="list-style-type: none"> <li>The malicious nodes are not detected by combining with machine learning approaches.</li> <li>The energy consumption is not computed in the transmission phase.</li> </ul>                  |
| Tamilarasi and Santhi [7]          | PSO                             | <ul style="list-style-type: none"> <li>It returns better network lifetime and energy consumption.</li> <li>It enhances the security of the network.</li> </ul>                                                                     | <ul style="list-style-type: none"> <li>It does not function outside the out-of-band channel.</li> </ul>                                                                                                                           |
| Vo <i>et al.</i> [8]               | MLAMAN                          | <ul style="list-style-type: none"> <li>The management protocol offered the certification of node membership.</li> <li>The malicious nodes are prevented from entering the network using pretense and false information.</li> </ul> | <ul style="list-style-type: none"> <li>The security performance is not enhanced by means of TLS library.</li> <li>It does not consider the practical scenarios that involve size of the network and speed of the node.</li> </ul> |

### III. PROPOSED ARCHITECTURE DEVELOPED FOR DETECTION AND MITIGATION OF WORMHOLE ATTACKS IN IOT

#### A. Proposed Model

Security in IoT is the major field to safeguard the connected networks and devices in the IoT environment. From the literature, diverse studies show that IoT networks are facing multiple security limitations in terms of attacks such as sybil attack, acknowledgement spoofing attack, sinkhole attack, HELLO flood attack and worm hole attack, etc. The major security threat in IoT is wormhole attack that affects routing protocols with the interruption by diverse attacks. IoT security is the significant and major problems in network sensors due to the inherent liabilities of physical devices. The security issues can be raised because of the sharing of same protocols by all

the nodes in the networks for transmission. Further, the nature of topology structure is dynamic and it is unguided one that creates the network as more vulnerable to attacks. In recent years, the implementation of attack detection and mitigation models has become increasingly complex, inspiring researchers to explore this area. Over the past few decades, a range of approaches have been developed for detecting and mitigating wormhole attacks in IoT, which have demonstrated efficacy in real-time applications. However, those methods are suffered from differentiating the correlation among affected nodes and bad networks and their behaviors, which should be solved. Therefore, there is a need to propose a detection and mitigation model against wormhole attacks, which is represented in Fig 1.

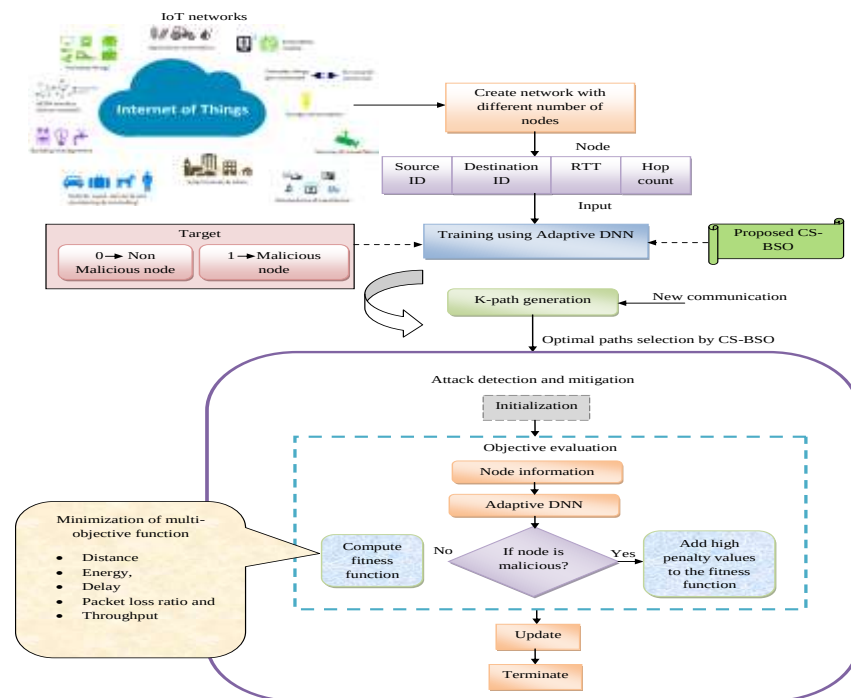


Fig. 1. Proposed Wormhole attack detection and prevention model in IoT

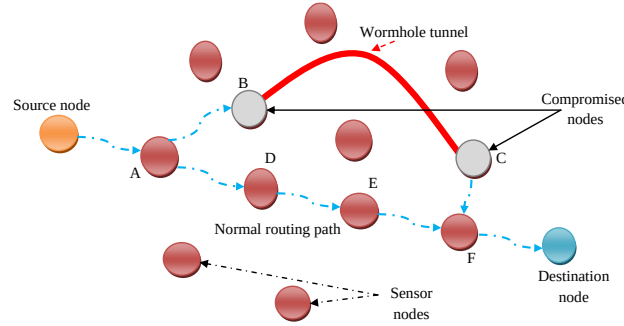
The proposed model considers different phases to detect and mitigate the wormhole attacks in IoT. They are (a) network creation, (b) network training, (c) wormhole attack detection and prevention. Initially, the network is created with specified count of nodes, and the parameters like “source ID, destination ID, RTT and hop-count” are extracted from each node. With this information, training is performed by ADNN by considering the parameters of nodes as input and the target is fixed with malicious nodes and non-malicious nodes as 1 and 0, respectively. The target is determined based on the abnormalities in RTT and hop count. Here, the training algorithm of Deep Neural Networks (DNN) is improved through the utilization of the proposed Cuckoo Search-Binary Bat Algorithm (CS-BSO). This algorithm optimizes the weights in order to minimize the discrepancy between the actual and measured outcomes. In the beginning stages of a new communication, the k-paths are initialized randomly. The k-path is generated based on neighbour nodes with source and destination nodes. The optimal path selection by the proposed CS-BSO algorithm from the generated k-paths is the major

contribution in the proposed model. During this process, detection of wormhole attack and its mitigation happens. Once the initialization of paths is done, the objective evaluation is performed for each solution (paths). The detection mechanism is happened inside the objective evaluation, where each node is checked with the trained ADNN. In the event that a node is detected with a wormhole attack, the proposed approach involves the imposition of a high penalty on the fitness function in order to eliminate the affected path. The fitness function is tailored to paths containing normal nodes, and is designed to minimize a multi-objective function that takes into account constraints such as distance, energy consumption, delay, packet loss, and throughput. As a result, the paths ultimately selected by this approach are intended to effectively detect and prevent wormhole attacks within the network.

#### B. Description of Attack

In this attack, one or multiple malicious nodes are present, where these nodes create a tunnel or bridge among them for communication. These nodes are considered as attack nodes

*Wormhole Detection Scheme with Adaptive Deep Neural Network and Hybrid Multi-Objective Mitigation for Internet of Things* that can attacks other sensor nodes. In Fig. 2, the general wormhole structure is given.



**Fig. 2. Representation of Wormhole attack**

There are series of nodes in the network, in which a new tunnel is created by wormhole attackers. From the diagram, the node B directly transmits the packets to node C that forwards the packets to node F. Thus, there can be early arrival of the packet at destination node that can create a network traffic, which results in crash. Here, the faster communication is achieved in the network because of the direct relationship among attackers and other nodes. The wormhole nodes or compromised nodes transmit faster than other nodes. The created tunnel can be used to drop or control the packets or data and for traffic analysis. This type of attacks is established by not revealing their identities.

#### IV. DETECTION OF WORMHOLE ATTACKS IN IOT USING ADAPTIVE DNN MODEL

##### A. Network Training

The network is trained by considering the input as node with parameters like “Source ID, Destination ID, RTT and Hop-Count”. These are used for the transmission of data in secured manner.

**Source ID:** It represents the identity of source node.

**Destination ID:** It indicates the identity of destination node.

**RTT:** The RTT of the  $rt^{th}$  route is described as the “the difference between the transmission time  $tr_{tx}$  of RREQ from source and reception time  $tr_{rx}$  of RREQ from the destination”, which is formulated in Eq. (1).

$$RTT_{rt} = tr_{rx} - tr_{tx} \quad (1)$$

**Hop count:** refers to the number of intermediary nodes that participate in the routing path between a source and destination. The calculation of hop count does not require any additional measurements.

##### B. Adaptive DNN-based Wormhole attack detection

The proposed approach employs the use of ADNN for the detection of wormhole attacks, whereby the weights of the DNN are optimized using the CS-BSO algorithm proposed herein. This ADNN is utilized for training the detection and mitigation model, whereby the input comprises node structure, while the target comprises malicious and non-malicious nodes, represented as 1 and 0, respectively. The DNN framework

comprises three key layers, namely the input layer, hidden layers, and output layer, along with their respective weights. The hidden layers are implemented within the DNN to effectively learn the mapping correlation between the input and output data. To enhance the training speed of the DNN, two hidden layers are designed. The total number of nodes in the hidden layer is computed in accordance with Eq. (2).

$$nh = \sqrt{ni + no + cv} \quad (2)$$

Here, the terms  $nh$ ,  $no$ ,  $ni$  and  $cv$  indicate the number of hidden layer nodes, the number of output layer nodes, the number of input layer nodes and a constant value among [1,10], respectively. In the hidden layer of DNN, an activation function is inserted to enable the capability of non-linear fitness, in which the DNN considers a sigmoid as an activation function as given in Eq. (3).

$$SF = \frac{1}{1 + e^{-ip}} \quad (3)$$

In Eq. (3), term  $ip$  denotes the input data of the network, which is activated through  $MF$ .

$$MF = \text{sigm}(\omega_y ip + \alpha_y) \quad (4)$$

Here, the bias between output and hidden layer is denoted as  $\alpha$  and the weight matrix is represented as  $\omega$ . The loss form is formulated for a sample by considering a conceptually labeled data sample as  $(ip, l)$ .

$$SF(Wb_{sf}, no_{sf}; ip, l) = \frac{1}{2nn} \sum_{jn=1}^{nn} \|hj_{jn}(Wb_{sf}, no_{sf}; ip) - l_{jn}\|_2^2 \quad (5)$$

In Eq. (5), the number of neurons in the hidden layer and the subsets of biases is represented as  $nn$  and  $Wb_{sf}, no_{sf}$ , respectively. Here, the weight matrix  $\omega$  is optimized using the proposed CS-BSO algorithm. It is done to reduce the error distinction among the actual and measured outcome as given in Eq. (6).

$$ERR = \arg \text{Min}_{\{\omega\}} \sum_{er=1}^{Err(s)} |Er_{er} - \hat{Er}_{er}| \quad (6)$$

In Eq. (6), the actual and predicted outcomes are represented as  $Er_{er}$  and  $\hat{Er}_{er}$ , respectively.

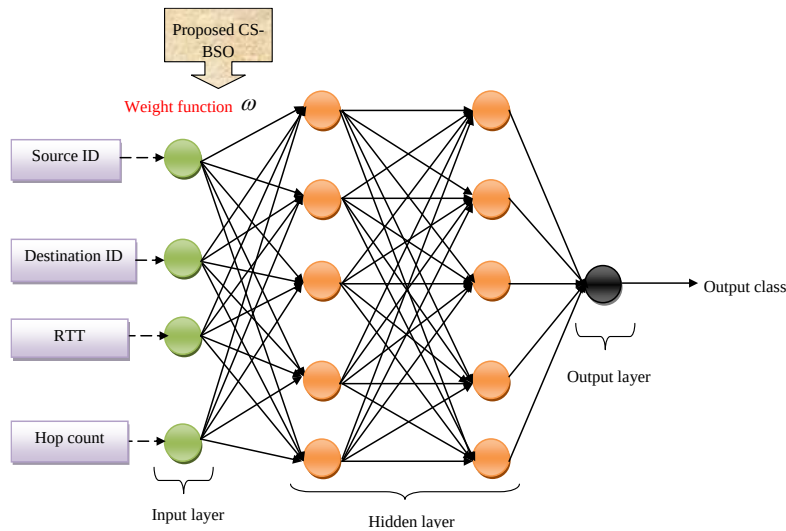


Fig. 3.Adaptive DNN structure

The ADNN that has been trained is utilized for optimal path selection, which aids in the detection and mitigation of wormhole attacks.

V. MITIGATION OF WORMHOLE ATTACKS IN IOT USING PROPOSED CS-BSO ALGORITHM

A. Optimal Path Generation

The proposed wormhole attack detection and mitigation model generates k-paths among source and destination nodes based on the neighbor nodes through a graph structure. The k-path is generated by considering the source node  $sn_{node}$  and destination

node  $dn_{node}$  together with IDs. Moreover, the optimal path is chosen from the diverse shortest route paths through proposed CS-BSO algorithm from the generated k-paths. It is considered as best route. The k-path routing process involves determining the minimum number of feasible shortest paths or bandwidth between two nodes within the communication area, while also ensuring that the minimum bandwidth value is met. To select the optimal path, the proposed CS-BSO with ADNN method is employed, which effectively detects and mitigates wormhole attacks within the network. Figure 4 illustrates the generation of the optimal path.

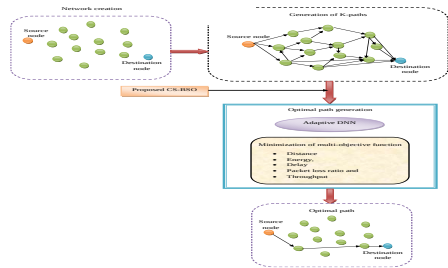


Fig. 4.Optimal path generation using CS-BSO algorithm

The optimal path generation is the major stage for attack detection and mitigation using ADNN and proposed CS-BSO algorithm. In this stage, the wormhole attack paths are

removed once it is detected. Fig. 5 provides the solution encoding for the optimal path selection.



Fig. 5.Solution encoding of optimal path generation using CS-BSO algorithm



Here, term  $k_{in}$  denotes the solution of k-paths, where a greater number of paths is generated with source & destination node, varied number of intermediate nodes. From  $k_{in}$ , the optimal paths are selected using the proposed CS-BSO algorithm, where the bounding limit of k-paths is from 1 to  $k_{in}$  and  $tn$  indicates the total number of paths generated.

## B. Objective Model for Mitigation

The primary aim of the proposed model is to select the most optimal paths  $k_{in}$  from  $k_{in}$  paths. From the  $k_{in}^{th}$  path, the data structure is determined, which is considered as the input of the ADNN. The optimal path generation is the major phase for the attack detection and mitigation, which is done by initializing the network with objective evaluation, where the node information is subjected to the trained ADNN to discover whether the node is malicious or not. In the event that a node is deemed to be malicious, a significant penalty is incorporated into the fitness values. Conversely, if the node is determined to be non-malicious, the fitness function is determined using a multi-objective function that takes into account parameters such as distance, energy, delay, packet loss, and throughput. These parameters represent the primary objectives of the developed attack detection and mitigation model.

Consider, term  $FF_{Di}$  denotes the function pertain to distance,  $FF_{En}$  indicates energy,  $FF_{De}$  represents delay,  $FF_{pkt}$  referred to as packet loss ratio and  $FF_{Trt}$  specifies throughput and the objective functions are formulated here.

$$ob_1 = FF_{Di}\alpha + (1-\alpha)\frac{1}{FF_{En}} \quad (7)$$

$$ob_2 = ob_1\beta + (1-\beta)\frac{1}{FF_{De}} \quad (8)$$

$$ob_3 = ob_2\gamma + (1-\gamma)\frac{1}{FF_{pkt}} \quad (9)$$

$$ob_4 = ob_3\delta + (1-\delta)\frac{1}{FF_{Trt}} \quad (10)$$

Here, the weighted factors are assigned as  $\alpha = 0.25$ ,  $\beta = 0.25$ ,  $\gamma = 0.25$  and  $\delta = 0.25$ . The final objective function is formulated in Eq. (11).

$$FF^* = \text{Min}(Ob_4) \quad (11)$$

The primary goal of the proposed attack prevention and detection model is to minimize the fitness function  $FF^*$ . Here, distance, packet loss and delay should attain minimum value and energy and throughput must be achieved specified rate for determining the optimal path. Here, the functions are formulated here.

**Distance:** The Euclidean distance among two nodes is compute here.

$$FF_{Di} = \sum_{jn=1}^{NJ} |x1_{jn} - y1_{jn}| \quad (12)$$

Here, the first node is indicated as  $x1_{jn}$  and the other node is referred to as  $y1_{jn}$ .

**Normalized Energy:** The formula presented in Eq. (13). defines the "average energy of nodes remaining available for broadcasting following energy dissipation.

$$FF_{En} = \frac{1}{Cn^*} \sum_{1}^{Cn^*} FF_{En}^{cn} \quad (13)$$

Here, term  $Cn$  indicates the considered nodes, which is  $cn = 1, 2, \dots, Cn^*$  and  $FF_{En}^{cn}$  is the power consumption of individual nodes.

**Latency or delay:** The term "latency" is defined as the time elapsed between a system's response and stimulation, or the delay between the source and target of a physical modification, as expressed in Eq. (14).

$$FF_{De} = Dr_{sn} - Dr_{dn} \quad (14)$$

In Eq. (14), The interval for receiving packets from a destination node is denoted as  $Dr_{dn}$  and The duration of packet transmission from the source node is represented as  $Dr_{sn}$ .

**Throughput:** It refers to "the amount of data that was successfully sent in a certain amount of time, via a particular communication link.", which is given in Eq. (15).

$$\text{Throughput} = \frac{Ct_{pkd} * sz_{pk} * 8}{tot_{sim}} \quad (15)$$

Here, term  $Ct_{pkd}$  represents the total number of packets that have been delivered,  $sz_{pk}$  is the size of the packet and  $tot_{sim}$  denotes the total duration of simulation.

**Packet loss ratio:** It refers to the failure of one or more sent packets to reach their intended destination, which is equated in Eq. (16).

$$PLR = \frac{trs_{Di}}{trs_{St}} \times 100 \quad (16)$$

Here, terms  $trs_{St}$  and  $trs_{Di}$  denotes data received and transmitted, respectively.

## C. Proposed CS-BSO

The developed attack detection and mitigation model proposes a new algorithm called CS-BSO algorithm. It is inspired from BSA [26] and CSA [27] for optimizing the weight of DNN and selection of optimal paths from the generated k-paths. BSA [26] is motivated by bird social interactions and behaviors like as "foraging behaviour, vigilance behaviour, and flight activity" in order to obtain food and increase their survival rate.. Consider, the number of birds in the swarm or virtual birds is given as  $NS$  with position as  $p_j^i = (p_{j,1}^i, p_{j,2}^i, \dots, p_{j,K}^i)$  for  $j^{th}$  bird at time  $i$ , where  $j \in [1, 2, \dots, NS]$  and  $i \in [1, 2, \dots, I]$ .

**Foraging behavior:** It is formulated in Eq. (17). Each bird searches for food based on this behavior with individual and swarm experiences.

$$p_{j,m}^{i+1} = p_{j,m}^i + (p_{j,m}^i - p_{j,m}^i) \times CE \times rand(0,1) + (q_m - p_{j,m}^i) \times SC \times rand(0,1) \quad (17)$$

Here, the positive numbers like social accelerated coefficients and cognitive accelerated coefficients are denoted as  $SC$  and  $CE$ , respectively, where  $SC = 1$ . The term  $p_{j,m}$  indicates the best historical position at  $m^{th}$  coordinate



of  $j^{th}$  bird, the independent uniformly distributed numbers in the range of (0, 1) is given as  $rand(0,1)$  and the best historical position shared through the swarm is termed as  $q_m$  at  $m^{th}$  coordinate. When a uniform random number is in (0,1) is lesser than  $\rho$  ( $\rho \in (0,1)$ ), the bird will search for food otherwise, the bird will persist on vigilance behavior. Here  $\rho$  is a constant value.

**Vigilance behavior:** The birds can move towards the centre of the birds swarm, and their movements are formulated in Eq. (18).

$$p_{j,m}^{i+1} = p_{j,m}^i + B_1 (mean_m - p_{j,m}^i) \times rand(0,1) + B_2 (q_{x,m} - p_{j,m}^i) \times rand(-1,1) \quad (18)$$

$$B_1 = b1 \times \exp\left(-\frac{PF_j}{SF + \varepsilon} \times NS\right) \quad (19)$$

$$B_2 = b2 \times \exp\left[\left(\frac{PF_j - PF_x}{|PF_j - PF_x| + \varepsilon}\right) \frac{PF_j}{SF + \varepsilon} \times NS\right] \quad (20)$$

Here, the positive constants are referred to as  $b1$  and  $b2$  in  $[0,2]$ , the mean position of the entire swarm is given as  $mean_m$  at  $m^{th}$  coordinate, a positive integer is denoted as  $x$  in range of  $[1, NS]$  with  $x \neq j$ . The fitness value related to the best historical position is given as  $PF_j$  for  $j^{th}$  bird, the sum of  $PF_j$  is indicated as  $SF$  and the zero division error is avoided by using  $\varepsilon$ .

**Flight behavior:** There are two types of birds in swarm that are producers and scroungers, where the producers indicate energetic searching of food while scroungers will follow a producer to search for food at random manner. These positions are updated using Eq. (21) and Eq. (22).

$$p_{j,m}^{i+1} = p_{j,m}^i \times randns(0,1) \times p_{j,m}^i \quad (21)$$

$$p_{j,m}^{i+1} = p_{j,m}^i + (p_{x,m}^i - p_{j,m}^i) \times LE \times p_{j,m}^i \quad (22)$$

Here, the scrounger learning efficiency is given as  $LE$  that is in  $LE \in [0,2]$  used for following the producer. This BSO algorithm has the ability for solving the extreme problems, good global search ability and good optimization precision. Though, it has diverse challenges such as premature convergence for high-dimensional functions, local optimal issues, and its population multiplicity is inadequate and it can easily fall into local extrema. These limitations have motivated the researchers for adapting a hybridization concept with CSA. CSA has diverse features like finding of optimal solutions is easier and simplicity of implementation. The CSA is inspired the researchers because of the intelligent behavior of stealing the foods and their cleverness that also use multiple strategies for stealing their foods. At initial stage, the parameter termed awareness probability ( $AP$ ) and population of crows is initialized. The position  $p^{i,j+1}$  of the search space is formulated in Eq. (23).

$$p^{i,j+1} = p^{i,j} + rn_i \times fl^{i,j} \times (hp^{i,j} - p^{i,j}) \quad (23)$$

Here, term  $p^{i,j}$  denotes the position of  $i^{th}$  crow at  $j^{th}$  iteration,  $fl^{i,j}$  specifies the flight length of crow  $i$  at  $j^{th}$  iteration, the position of hiding place is denoted as  $hp^{i,j}$  for crow  $i$ ,  $rn_i$  indicates the random number in the range of 0 and 1, and the awareness probability of crow  $i$  at  $j^{th}$  iteration is referred to as  $AP$ . Thus, a new CS-BSO algorithm is developed based on the random parameter  $\rho$  of BSO and  $AP$  of CSA. If  $rand(0,1) > \rho(j)$ , then the position is updated using the foraging behaviour of BSO algorithm; else a condition is defined. Here, if  $rand(0,1) \leq AP$ , then position is updated using CSA; otherwise position is updated by vigilance behavior of BSO algorithm. Algorithm 1 contains the pseudo code for the implemented CS-BSO algorithm.

#### Algorithm 1: Proposed CS-BSO

```

Initialize the population and parameters like the maximum number of iteration as  $MI$  and the frequency of birds' flight behaviors is indicated as  $FB$ 
Determine best solutions and fitness values
While ( $i < MI$ )
    if ( $i \% FB \neq 0$ )
        for  $j = 1 : NS$ 
            if  $rand(0,1) < \rho(j)$ 
                Update position based on foraging behavior of BSO using Eq. (17)
            else
                if  $rand(0,1) > AP$ 
                    Update position using CSA by Eq. (23)
                else
                    Update position based on vigilance behavior of BSO using Eq. (18)
                end
            end
        end
        Separate the swarm into producers and scroungers
    for  $j = 1 : NS$ 

```

```

        if  $j$  is considered as producer
            Update Positions Using Eq. (21)
        else
            Update Positions Using Eq. (22)
        end
    end for
end if
end for
end
Determine new solutions
Compute best solutions
 $i = i + 1$ 
end while
    
```

The hybridization concept attains better performance in diverse literature works [31]. It is developed by adapting two or more principles from two different approaches. It aims to solve some specific search problems with promising solutions. Also, this

hybrid theory offers better convergence than other meta-heuristic approaches. The flowchart of the developed CS-BSO algorithm is given in Fig 6.

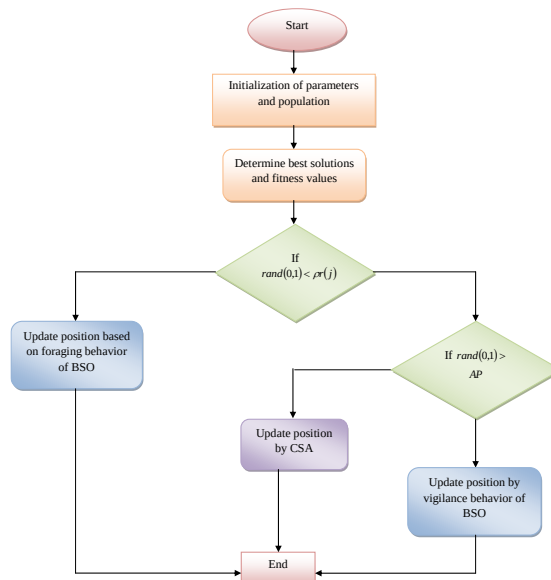


Fig. 6. Flowchart of the proposed CS-BSO algorithm

Fig. 7.

## VI. RESULTS AND DISCUSSIONS

### A. Experimental Setup

The developed wormhole attack detection and prevention was experimented in MATLAB 2019a, and the results were analysed. The experimentation was done by considering the four networks like Network 1 consists of total number of nodes as 35 nodes, network 2 includes 50 nodes, network 3 has 65 nodes and 4<sup>th</sup> network consists of 80 nodes, where each network has 800 rounds. The experiment was carried out with the maximum number of iterations and population size set to ten. The suggested model's performance was compared to that of standard models in terms of convergence analysis, as well as analysis in terms of normalized energy, latency, packet loss ratio, shortest path, and throughput. It was compared with diverse conventional models such as Bypass-Linked Attacker Update-based ROA (BAU-ROA) [29], D-DHOA [28].

### B. Performance measures

The performance of the proposed attack detection and mitigation model is analyzed by considering the performance

measures. Here,  $po^{tr}$ ,  $ng^{tr}$ ,  $po^{fa}$ ,  $ng^{fa}$  refer to the “true positives, true negatives, false positives, and false negatives”, respectively.

(a) **Accuracy:** “ratio of perfectly anticipated observations to total observations”.

$$Ary = \frac{(po^{tr} + ng^{tr})}{(po^{tr} + ng^{tr} + po^{fa} + ng^{fa})} \quad (24)$$

(b) **Precision:** “the ratio of exactly anticipated positive observations to the total number of positively predicted observations”.

$$Pr = \frac{po^{tr}}{po^{tr} + po^{fa}} \quad (25)$$

(c) **Specificity:** “the number of true negatives, which are determined precisely”.

$$Spy = \frac{ng^{tr}}{ng^{fa}} \quad (26)$$

(d) **Sensitivity**: “the number of true positives, which are recognized exactly”.

$$Sy = \frac{po^{tr}}{po^{tr} + ng^{fa}} \quad (27)$$

(e) **FPR**: “the ratio of count of false positive predictions to the entire count of negative predictions”.

$$FPR = \frac{po^{fa}}{po^{fa} + ng^{tr}} \quad (28)$$

(f) **F1 score**: “harmonic mean between precision and recall. It is used as a statistical measure to rate performance”.

$$F1score = \frac{Sy \cdot Pr}{Pr + Sy} \quad (29)$$

(g) **FNR**: “the proportion of positives which yield negative test outcomes with the test”.

$$FNR = \frac{ng^{fa}}{ng^{tr} + po^{tr}} \quad (30)$$

(h) **FDR**: “the number of false positives in all of the rejected hypotheses”.

$$FDR = \frac{po^{fa}}{po^{fa} + po^{tr}} \quad (31)$$

(i) **MCC**: “correlation coefficient computed by four values”.

$$MCC = \frac{po^{tr} \times ng^{tr} - po^{fa} \times ng^{fa}}{\sqrt{(po^{tr} + po^{fa})(po^{tr} + ng^{fa})(ng^{tr} + po^{fa})(po^{tr} + ng^{fa})}} \quad (32)$$

(j) **NPV**: “probability that subjects with a negative screening test truly don't have the disease”.

$$NPV = \frac{ng^{fa}}{ng^{fa} + ng^{tr}} \quad (33)$$

### C. Convergence Analysis

Fig.7 presents the convergence analysis of both the proposed and conventional models for mitigating wormhole attacks. At the initial iteration, the proposed CS-BSO+ADNN attain minimal cost function than other approaches. It is maintained at all the iterations when using the suggested CS-BSO-based DNN. At iteration 2, the cost function of the proposed CS-BSO+ADNN is followed by D-DHOA+ADNN, BAU-ROA+ADNN, CSA+ADNN and BSA+ADNN. At 3<sup>rd</sup> iteration, the cost function of the proposed CS-BSO+ADNN is 23% enhanced than D-DHOA+ADNN, 37.5% enhanced than BAU-ROA+ADNN, 42.8% enhanced than CSA+ADNN and 45.9% enhanced than BSA+ADNN. Therefore, it is observed that the cost function of the developed attack detection and mitigation model using CS-BSO+ADNN is outperformed the traditional approaches.

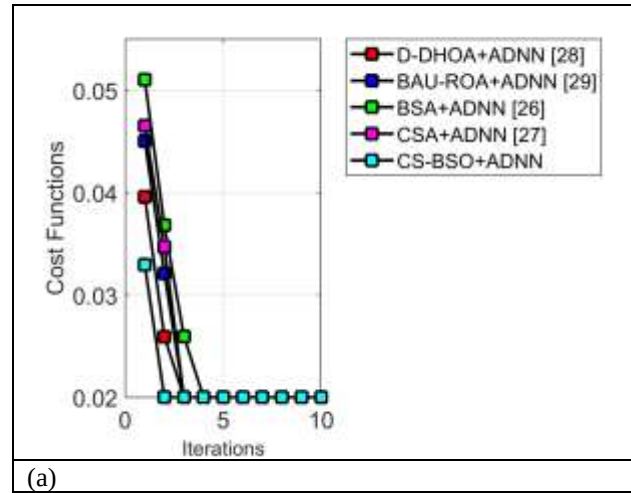


Fig. 8. Convergence analysis of the implemented and traditional models for wormhole detection and mitigation in IoT

### D. Performance analysis of Attack Detection for different networks

The performance evaluation of the proposed attack detection model has been conducted on four distinct networks, as presented in Tables II through V. The results demonstrate that the proposed attack detection and mitigation model outperforms conventional meta-heuristic techniques for all the networks, thereby enhancing the overall performance of the system. The accuracy of the proposed CS-BSO-DNN is 3.3% superior to D-DHOA-DNN and BAU-ROA-DNN, 4.8% superior to CSA-DNN and 4% superior to BSA-DNN for network 1. Similarly, for network 1, the FNR of the developed CS-BSO-DNN is 22.9%, 25%, 30% and 28.7% minimized than D-DHOA-DNN, BAU-ROA-DNN, CSA-DNN and BSA-DNN, respectively. When considering the network 2, the accuracy of the developed CS-BSO-DNN is 2.8%, 3.2%, 2.9% and 3.3% better than D-DHOA-DNN, BAU-ROA-DNN, CSA-

DNN and BSA-DNN, respectively. Also, for network 2, the FPR of the implemented CS-BSO-DNN is 33%, 29%, 33% and 37% reduced than D-DHOA-DNN, BAU-ROA-DNN, CSA-DNN and BSA-DNN, respectively. Likewise, on considering network 3, accuracy of the proposed CS-BSO-DNN is 2%, 4%, 2.9% and 4% enhanced than D-DHOA-DNN, BAU-ROA-DNN, CSA-DNN and BSA-DNN, respectively. The NPV of the suggested CS-BSO-DNN is 0.58% better than D-DHOA-DNN, 0.35% better than BAU-ROA-DNN, 0.73% better than CSA-DNN and 0.76% better than BSA-DNN for network 3. Correspondingly, the accuracy of the implemented CS-BSO-DNN for network 4 is 3%, 2.3%, 3% and 3.7% improved than D-DHOA-DNN, BAU-ROA-DNN, CSA-DNN and BSA-DNN, respectively. The specificity of the considered CS-BSO-DNN is 0.6%, 0.7%, 0.74% and 0.89% progressed than D-DHOA-DNN, BAU-ROA-DNN, CSA-DNN and BSA-DNN, respectively for network 4. The performance analysis of

*Wormhole Detection Scheme with Adaptive Deep Neural Network and Hybrid Multi-Objective Mitigation for Internet of Things*  
the developed attack detection and mitigation model using CS- techniques.  
BSO-DNN indicates superior results compared to alternative

**TABLE II. OVERALL PERFORMANCE ANALYSIS NETWORK 1 INTERMS OF WORMHOLE ATTACK DETECTION**

| Measures    | D-DHOA-DNN [28] | BAU-ROA-DNN [29] | BSA-DNN [26] | CSA-DNN [27] | CS-BSO-DNN |
|-------------|-----------------|------------------|--------------|--------------|------------|
| Accuracy    | 0.935           | 0.935            | 0.92125      | 0.92875      | 0.96625    |
| Sensitivity | 0.32727         | 0.30769          | 0.25806      | 0.27273      | 0.48148    |
| Specificity | 0.97987         | 0.97861          | 0.97696      | 0.97718      | 0.98318    |
| Precision   | 0.54545         | 0.5              | 0.48485      | 0.46875      | 0.5        |
| FPR         | 0.020134        | 0.02139          | 0.023035     | 0.022819     | 0.016818   |
| FNR         | 0.67273         | 0.69231          | 0.74194      | 0.72727      | 0.51852    |
| NPV         | 0.97987         | 0.97861          | 0.97696      | 0.97718      | 0.98318    |
| FDR         | 0.45455         | 0.5              | 0.51515      | 0.53125      | 0.5        |
| F1-Score    | 0.40909         | 0.38095          | 0.33684      | 0.34483      | 0.49057    |
| MCC         | 0.39079         | 0.36018          | 0.316        | 0.32269      | 0.47321    |

**TABLE III. OVERALL PERFORMANCE ANALYSIS NETWORK 2 INTERMS OF WORMHOLE ATTACK DETECTION**

| Measures    | D-DHOA-DNN [28] | BAU-ROA-DNN [29] | BSA-DNN [26] | CSA-DNN [27] | CS-BSO-DNN |
|-------------|-----------------|------------------|--------------|--------------|------------|
| Accuracy    | 0.92875         | 0.925            | 0.9275       | 0.92375      | 0.955      |
| Sensitivity | 0.2931          | 0.29688          | 0.3          | 0.26667      | 0.34211    |
| Specificity | 0.97844         | 0.97962          | 0.97838      | 0.97703      | 0.98556    |
| Precision   | 0.51515         | 0.55882          | 0.52941      | 0.48485      | 0.54167    |
| FPR         | 0.021563        | 0.02038          | 0.021622     | 0.022973     | 0.014436   |
| FNR         | 0.7069          | 0.70313          | 0.7          | 0.73333      | 0.65789    |
| NPV         | 0.97844         | 0.97962          | 0.97838      | 0.97703      | 0.98556    |
| FDR         | 0.48485         | 0.44118          | 0.47059      | 0.51515      | 0.45833    |
| F1-Score    | 0.37363         | 0.38776          | 0.38298      | 0.34409      | 0.41935    |
| MCC         | 0.35408         | 0.37184          | 0.36347      | 0.32276      | 0.40857    |

**TABLE IV. OVERALL PERFORMANCE ANALYSIS NETWORK 3 INTERMS OF WORMHOLE ATTACK DETECTION**

| Measures    | D-DHOA-DNN [28] | BAU-ROA-DNN [29] | BSA-DNN [26] | CSA-DNN [27] | CS-BSO-DNN |
|-------------|-----------------|------------------|--------------|--------------|------------|
| Accuracy    | 0.94375         | 0.925            | 0.93625      | 0.925        | 0.96375    |
| Sensitivity | 0.3913          | 0.2623           | 0.34         | 0.3          | 0.42857    |
| Specificity | 0.97745         | 0.9797           | 0.976        | 0.97568      | 0.98316    |
| Precision   | 0.51429         | 0.51613          | 0.48571      | 0.5          | 0.48       |
| FPR         | 0.022546        | 0.020298         | 0.024        | 0.024324     | 0.016839   |
| FNR         | 0.6087          | 0.7377           | 0.66         | 0.7          | 0.57143    |
| NPV         | 0.97745         | 0.9797           | 0.976        | 0.97568      | 0.98316    |
| FDR         | 0.48571         | 0.48387          | 0.51429      | 0.5          | 0.52       |
| F1-Score    | 0.44444         | 0.34783          | 0.4          | 0.375        | 0.45283    |
| MCC         | 0.4197          | 0.33278          | 0.37397      | 0.35026      | 0.43489    |

**TABLE V. OVERALL PERFORMANCE ANALYSIS NETWORK 4 INTERMS OF WORMHOLE ATTACK DETECTION**

| Measures    | D-DHOA-DNN [28] | BAU-ROA-DNN [29] | BSA-DNN [26] | CSA-DNN [27] | CS-BSO-DNN |
|-------------|-----------------|------------------|--------------|--------------|------------|
| Accuracy    | 0.9325          | 0.93875          | 0.9325       | 0.92625      | 0.96125    |
| Sensitivity | 0.34483         | 0.33333          | 0.33929      | 0.31667      | 0.34483    |
| Specificity | 0.97844         | 0.97739          | 0.97715      | 0.97568      | 0.98444    |
| Precision   | 0.55556         | 0.48485          | 0.52778      | 0.51351      | 0.45455    |
| FPR         | 0.021563        | 0.022606         | 0.022849     | 0.024324     | 0.015564   |
| FNR         | 0.65517         | 0.66667          | 0.66071      | 0.68333      | 0.65517    |
| NPV         | 0.97844         | 0.97739          | 0.97715      | 0.97568      | 0.98444    |
| FDR         | 0.44444         | 0.51515          | 0.47222      | 0.48649      | 0.54545    |
| F1-Score    | 0.42553         | 0.39506          | 0.41304      | 0.39175      | 0.39216    |
| MCC         | 0.40437         | 0.37107          | 0.38947      | 0.36662      | 0.37633    |

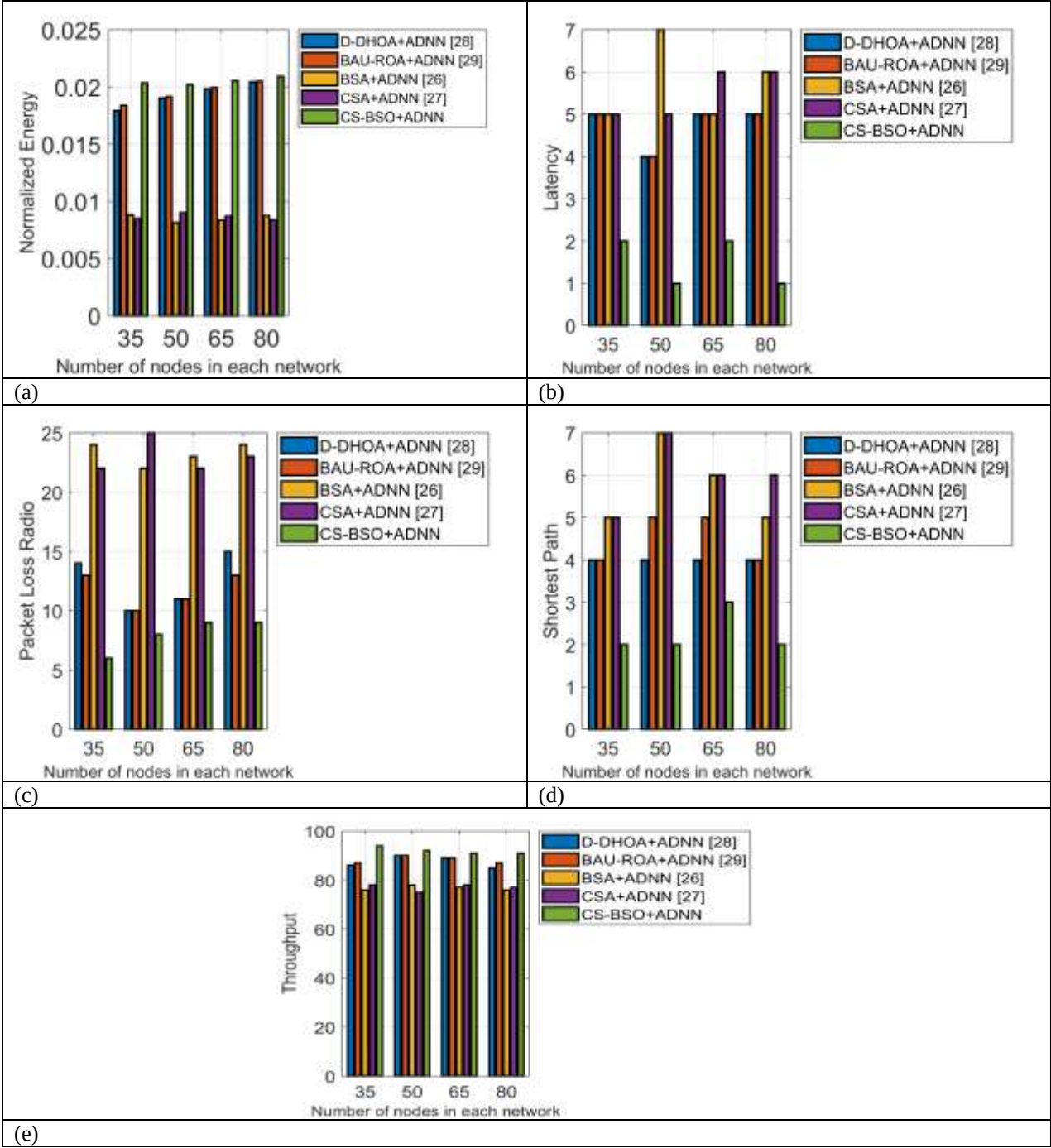
## E. Analysis for attack mitigation

The performance of the proposed attack mitigation using CS-BSO+ADNN algorithm is analyzed in Fig 8. The network's

performance is evaluated using various metrics while altering the number of nodes within it. The suggested CS-BSO+ADNN method has a greater efficiency in general than any other

algorithm. The normalized energy of the suggested CS-BSO+ADNN algorithm is 33%, 32.4%, 72% and 73% enhanced than D-DHOA+ADNN, BAU-ROA+ADNN, CSA+ADNN and BSA+ADNN, respectively when taking into account the number of nodes being 35. Likewise, the latency of the developed CS-BSO+ADNN algorithm is 75% minimized than D-DHOA+ADNN and BAU-ROA+ADNN, respectively and 85.7% minimized than CSA+ADNN and 80% minimized than BSA+ADNN by using a value of 50 for the node count. On taking the number of nodes as 65, the packet loss ratio of the developed CS-BSO+ADNN algorithm is 18% reduced than D-DHOA+ADNN and BAU-ROA+ADNN, respectively and 60% and 59% reduced than

CSA+ADNN and BSA+ADNN , respectively. The shortest path communication of the implemented CS-BSO+ADNN algorithm is 50% decreased than D-DHOA+ADNN, 50% decreased than BAU-ROA+ADNN, 60% decreased than CSA+ADNN and 66% decreased than BSA+ADNN, respectively at number of nodes as 80. The throughput of the developed CS-BSO+ADNN algorithm is 7%, 2%, 20% and 16.8% enhanced than D-DHOA+ADNN, BAU-ROA+ADNN, CSA+ADNN and BSA+ADNN, respectively at number of nodes as 80. Hence, it is confirmed that the performance of the developed attack mitigation model using CS-BSO+ADNN establishes the superior performance than other methods.

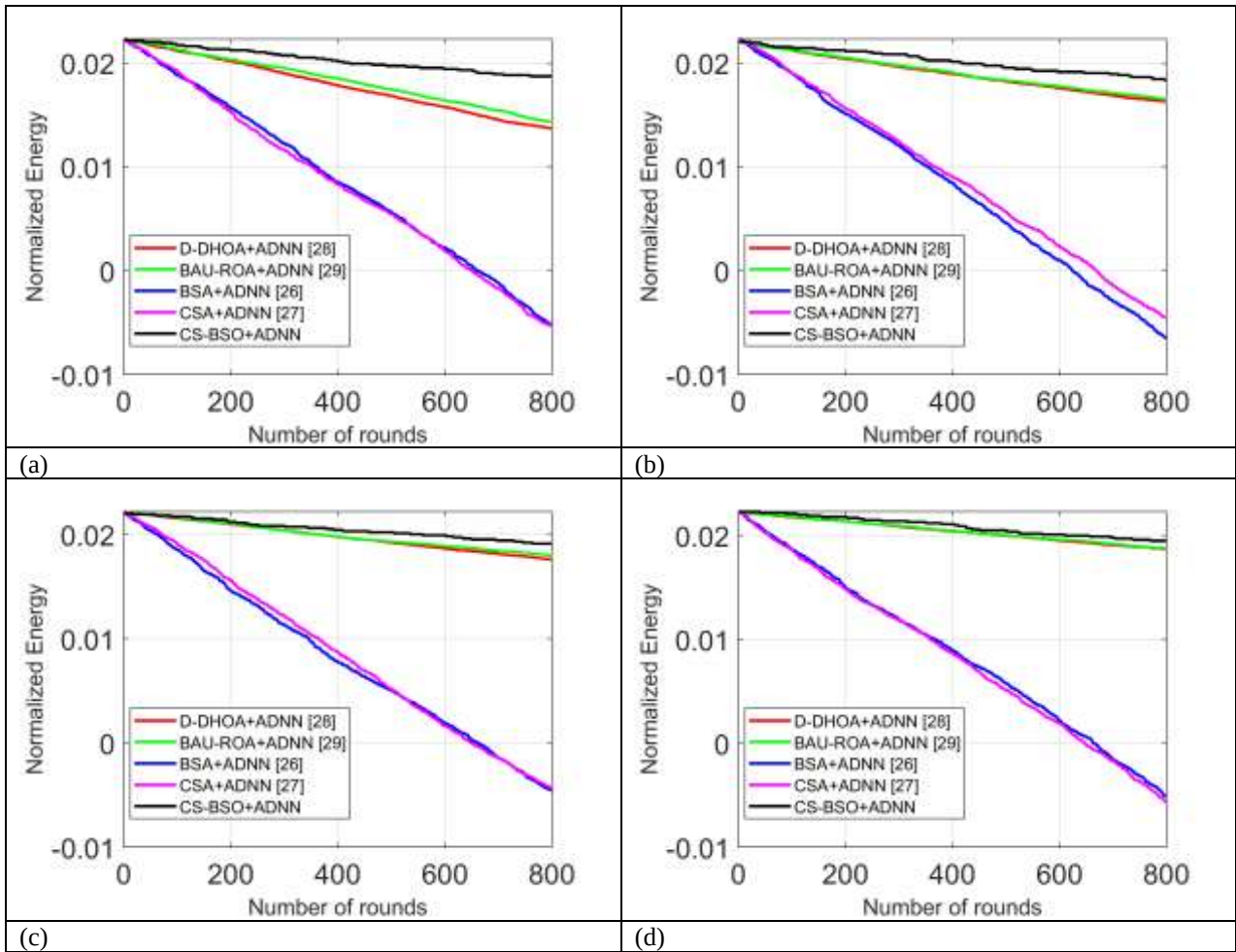


**Fig. 9.** Performance analysis of the developed and traditional models for wormhole attack mitigation in IoT in terms of (a) Normalized energy, (b) Latency, (c) Packet Loss Ratio, (d) Shortest path and (e) Throughput

**F. Analysis on Normalized energy for different networks**

The normalized energy is analyzed by varying the number of rounds from 0 to 800, which is given for 4 networks as given in Fig 9. It is clearly seen that the normalized energy of the suggested CS-BSO+ADNN is improved than other approaches for all the networks. In initial rounds, all the algorithms are performing well, but when the number of rounds is increased, the proposed detection model using CS-BSO+ADNN

establishes better performance than other algorithms in all four networks. For network 1, at round 800, the normalized energy of the developed CS-BSO+ADNN is 20% better than D-DHOA+ADNN and 12.5% better than BAU-ROA+ADNN. Likewise, the normalized energy of the suggested CS-BSO+ADNN is enhanced than other approaches for all the networks.



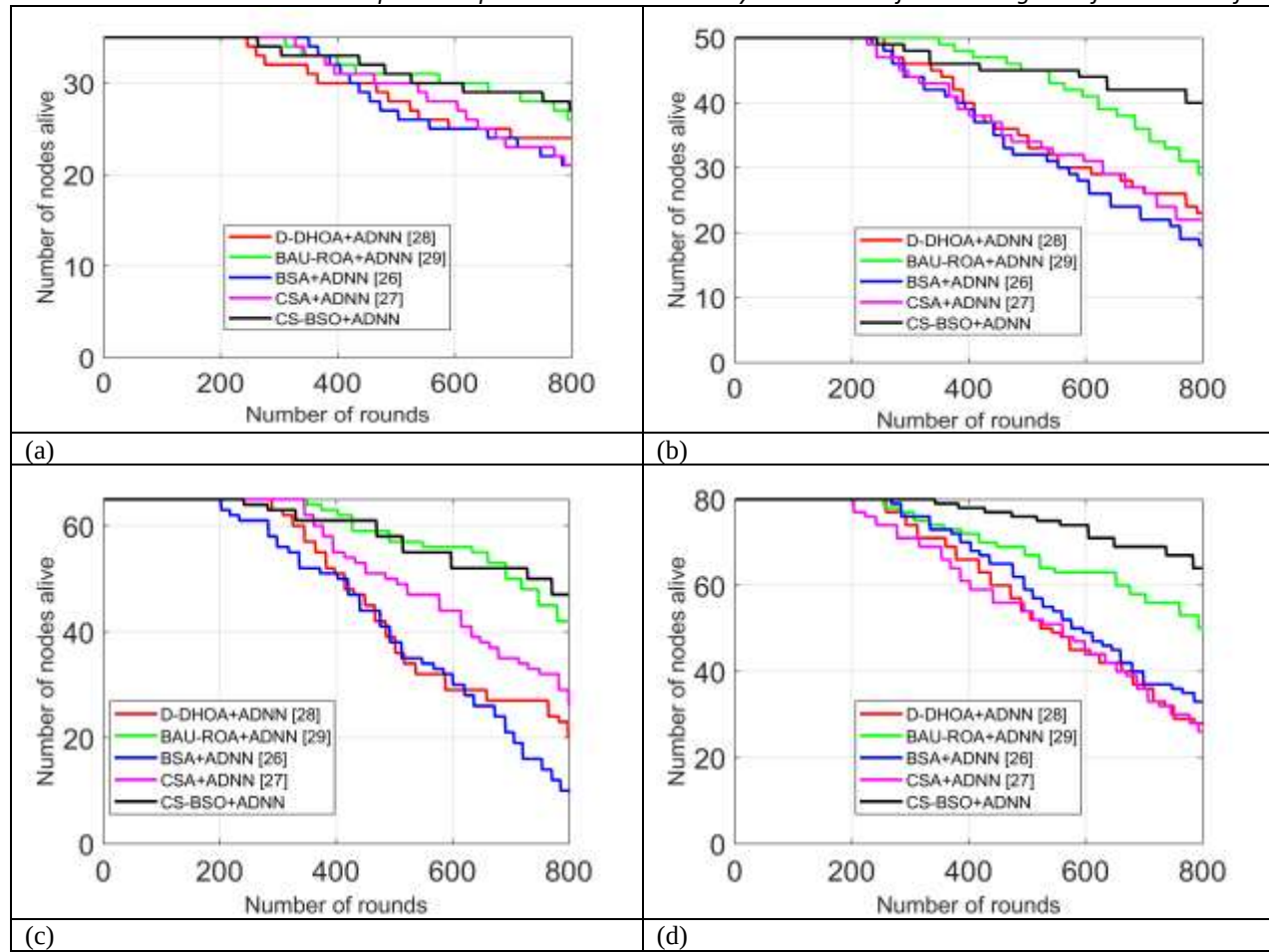
**Fig. 10.** Analysis on Normalized energy of the implemented and traditional models for wormhole attack mitigation in IoT for (a) Network 1, (b) Network 2, (c) Network 3 and (d) Network 4

**G. Analysis on alive nodes**

The number of alive nodes is analyzed for four networks when increasing the number of nodes, which is shown in Fig 10. In network 1, from 0 to 250 rounds, the proposed CS-BSO+ADNN get more number of alive nodes than other algorithms. However, after 250<sup>th</sup> rounds, the other algorithms like BSA+ADNN, CSA+ADNN and BAU-ROA+ADNN achieves maximum number of nodes than the proposed CS-BSO+ADNN. Further, at 800<sup>th</sup> round, the maximum number of alive nodes is achieved by the proposed CS-BSO+ADNN for all the networks than other conventional approaches. At 800<sup>th</sup>

round, for 2<sup>nd</sup> network, the number of alive nodes achieved by the suggested CS-BSO+ADNN is 66%, 37.9%, 55% and 81.8% superior to D-DHOA+ADNN , BAU-ROA+ADNN , CSA+ADNN and BSA+ADNN , respectively. Similarly, for the network 3, the number of alive nodes is attained as 48 at 800<sup>th</sup> round, which is far better than traditional algorithms. Likewise, the number of alive nodes achieved by the developed CS-BSO+ADNN is better than other techniques from the initial number of rounds to 800<sup>th</sup> rounds. Therefore, the number of alive nodes attained by the proposed CS-BSO+ADNN algorithm is more than other approaches.





**Fig. 11. Analysis on alive nodes for the developed and traditional models for wormhole attack mitigation in terms of (a) Network 1, (b) Network 2, (c) Network 3 and (d) Network 4**

## VII. CONCLUSION

This paper presents a novel approach for securing IoT systems by detecting and mitigating wormhole attacks. The proposed solution employs an artificial deep neural network (ADNN) in conjunction with a new algorithm called CS-BSO for detecting the attacks during the objective evaluation phase. Additionally, the proposed algorithm facilitates the mitigation process by optimizing path selection based on minimizing a multi-objective function derived from network parameters such as distance, energy, delay or latency, packet loss ratio, and throughput.

The simulation results demonstrate that the proposed CS-BSO-DNN-based detection algorithm outperforms other conventional models. Specifically, the accuracy of the proposed algorithm was found to be 3.3% better than D-DHOA-DNN and BAU-ROA-DNN, 4.8% better than CSA-DNN and 4% better than BSA-DNN for network 1. Moreover, the proposed CS-BSO+ADNN algorithm yielded a higher throughput for the network with 35 nodes, achieving enhancements of 7%, 2%, 20%, and 16.8% over D-DHOA+ADNN, BAU-ROA+ADNN, CSA+ADNN, and BSA+ADNN, respectively.

In summary, the proposed approach for wormhole attack detection and mitigation based on ADNN and CS-BSO has demonstrated superior performance compared to conventional

models, and has the potential to enhance the security of IoT systems.

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